



Ferromagnetic and photocatalytic properties of CrCl_3 monolayer: a spin-polarized ab initio study

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Accepted: 25 June 2024 / Published online: 4 July 2024
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Abstract

The current study presents a characterization of a novel CrCl_3 monolayer using spin-polarized density functional theory (DFT). With its monoclinic crystal structure, the CrCl_3 monolayer displays ferromagnetic properties due to its spin-dependent electronic behaviour. The electronic band gaps of 2.06 eV and 4.63 eV for spin-up and spin-down configurations, respectively, suggest semiconducting character. The magnetic ground state likely arises from strong hybridization between *Cr-d* and *Cl-p* states, resulting in an energetically stable ferromagnetic state. Additionally, calculations were conducted to evaluate the photocatalytic activity of the CrCl_3 monolayer. The Gibbs free energy for hydrogen adsorption was determined to be 0.13 eV. To explore variations in photocatalytic activity through catalyst functionalization, the CrCl_3 monolayer was functionalized with fluorine (F) and bromine (Br) substitutions. Substantial differences were observed in the electronic band gaps for both F- and Br-substituted CrCl_3 monolayers. Particularly noteworthy was the remarkable enhancement in photocatalytic activity observed with the F-substituted CrCl_3 monolayer.

Keywords 2D materials · DFT · Ferromagnetism · Photocatalysis

1 Introduction

The global demand for energy continues to escalate in tandem with our modern way of life. Residences, industries, and transportation collectively consume a substantial volume of energy, estimated at 0.5 zettajoule (10^{21} joule), annually [1, 2]. To confront this challenge,

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we require an energy source that is both clean and sustainable while maximizing efficiency. Hydrogen fuel has garnered significant attention recently as a potential solution; however, existing production methods reliant on fossil fuels emit harmful pollutants [3–5]. Consequently, researchers are actively pursuing environmentally friendly approaches to hydrogen production.

One promising solution lies in photocatalytic water splitting utilizing semiconductors [6, 7]. Research indicates that a mere 80 min of sunlight reaching the Earth could satisfy our annual energy requirements [8]. This technique employs semiconductors to convert sunlight into chemical fuels, such as hydrogen, in a cost-effective and scalable manner. Semiconductors play a pivotal role in this process. When photons, or light particles, possessing adequate energy interact with the semiconductor, they stimulate electrons, leaving behind “holes.” These electron-hole pairs are indispensable for catalytic activity. While titanium dioxide (TiO_2) was initially employed for this purpose [9], ongoing research continually introduces new catalysts. Given that the majority of reactions occur on the catalyst’s surface, materials with a substantial surface area are highly desirable. Among these, two-dimensional (2D) materials exhibit particular promise due to their remarkable properties.

The remarkable properties of two-dimensional (2D) materials have captured the attention of researchers over the past decade [10, 11]. Compared to their bulk counterparts, 2D materials exhibit increased strength, elasticity, charge carrier mobility, high surface-to-volume ratio, and numerous active adsorption sites [12]. Along with these extraordinary virtues, the reduced dimensionality makes 2D materials highly sought-after for various applications, including gas sensing, photocatalysis, hydrogen storage, metal-ion batteries, and spintronic devices [13–15]. For energy-efficient data storage, 2D materials can be used to make compact memory devices, providing a promising alternative to complementary metal-oxide semiconductor (CMOS) devices [16]. Furthermore, due to their numerous adsorption sites, 2D materials are excellent catalysts for the hydrogen evolution reaction (HER) [17, 18].

Chromium trichloride (CrCl_3) is a layered magnetic material that has garnered attention for its intriguing properties in both bulk and monolayer forms. In its bulk form, CrCl_3 crystallizes in a monoclinic structure where layers are held together by van der Waals forces, making it possible to exfoliate down to monolayers. Bulk CrCl_3 exhibits antiferromagnetic ordering below its Néel temperature of approximately 17 K [19]. However, when thinned down to a monolayer, CrCl_3 demonstrates distinct magnetic properties, transitioning to a ferromagnetic order at low temperatures. This change in magnetic behaviour is attributed to the increased importance of two-dimensional interactions and the reduced coordination environment of chromium atoms in the monolayer [20]. Its 2D nature also provides a high surface area and abundant active sites, which are advantageous for catalytic applications such as the hydrogen evolution reaction (HER). These unique properties of CrCl_3 in its monolayer form open new avenues for research and potential applications in spintronics, catalysis, and energy storage.

Given these advantages, our study focuses on chromium trichloride (CrCl_3) monolayer, investigating its potential both as a spintronic device and as a catalyst for hydrogen evolution. In this work, density functional theory (DFT) based computations have been performed to investigate the structural properties of the CrCl_3 monolayer. We have calculated the electronic properties, including electronic band structure and projected and total density of states (PDOS and TDOS), incorporating spin polarization effects to elucidate the electronic and magnetic behaviour of the CrCl_3 monolayer. Additionally, we explored the

catalytic activity for hydrogen evolution and examined the effect of fluorine and bromine substitution on the catalytic performance of the CrCl_3 monolayer for hydrogen evolution.

2 Computational details

The first principles characterization of CrCl_3 monolayer was carried out using *SIESTA* package [21]. Norm conserving *Troullier-Martin* pseudopotentials were employed on the basis of generalized gradient approximation (GGA) including the functional suggested by *Perdew-Burke-Ernzerhof* (PBE) [22]. Double ζ -polarization, based on split-valence method, was employed to define numerical basis set. The kinetic energy cut off of 450 eV was kept and the *Brillouin* zone sampling was done using *Monkhorst-Pack* grid of $12 \times 12 \times 1$ [23]. A $3 \times 3 \times 1$ supercell was built, keeping periodicity along X and Y axes and a vacuum of 20 Å along z-direction, avoiding interlayer interactions. The geometrical optimization was done until the convergence of 0.01 eV/Å for force on every atom and 10^{-4} eV for total energy was achieved. To investigate the magnetic behaviour of CrCl_3 monolayer, spin polarized computations of electronic as well as optical properties were carried out. The electronic band structure was plotted along Γ -M-K- Γ direction.

3 Results and discussion

3.1 Structural properties

A group of researchers studied the electronic and magnetic properties of trichloride of chromium (CrCl_3) in its bulk form, having monoclinic as well as trigonal configurations [24]. In two-dimensional (2D) state, CrCl_3 monolayer attains trigonal structure with R-3 space group. The computed lattice parameters are $a=b=5.99$ Å and $\gamma=120^\circ$. The length of the bond between chromium atom and chlorine atom is 2.35 Å, whereas the results obtained for bulk trigonal CrCl_3 by Wang et al., show a bond length of 2.36 Å [24]. As shown in Fig. 1, the unit cell of CrCl_3 monolayer consists of two formula units. The system has three layers,

Fig. 1 Top and side view of CrCl_3 monolayer

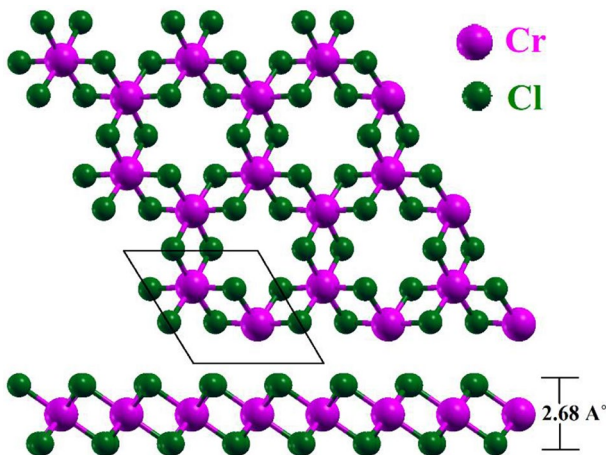


Fig. 2 Electronic band structure and projected and total density of states of states of pristine CrCl_3 monolayer. The Fermi level is shifted to 0 eV

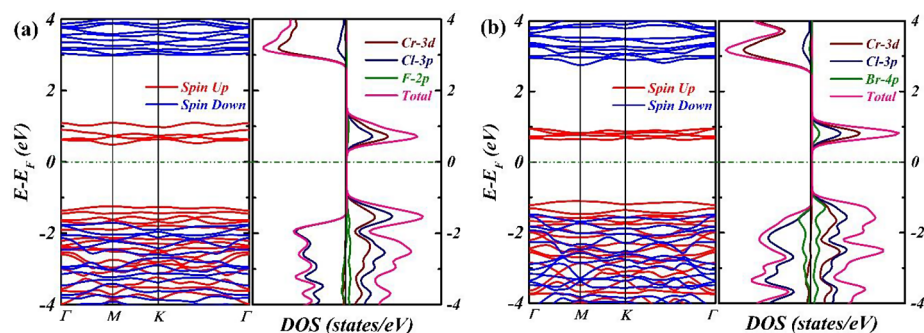
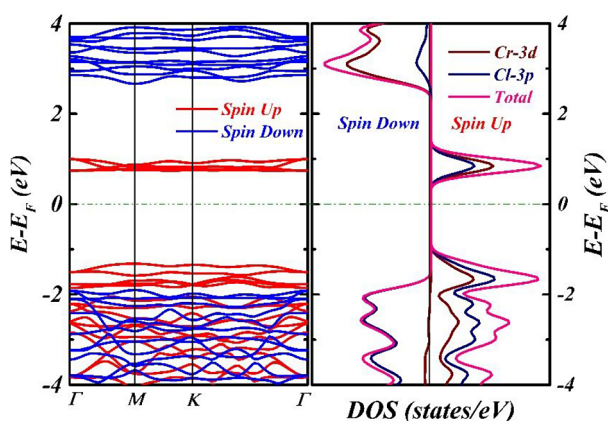


Fig. 3 Electronic band structure and projected and total density of states of (a) F-CrCl_3 and (b) Br-CrCl_3 monolayer. The Fermi level is shifted to 0 eV

Cl-Cr-Cl , stacked along Z-axis. Here, the stacking of triple layers is of the type A-Cr-B, where A and B are layers of Cl with different orientations. Each Cr atom, along with the six neighbouring Cl atoms, forms octahedral coordination. The thickness of the CrCl_3 monolayer is 2.68 Å.

3.2 Electronic properties

The electronic band structure with the inclusion of spin polarization, along with the projected and total density of states (DOS) are shown in Fig. 2. In the electronic band structure, red lines represent the contributions of spin up electrons and the blue lines represent the contributions of spin down electrons. For spin up contributions, the indirect electronic band gap of 2.06 eV. Here, the hole crystal momentum and electron crystal momentum are different. Also, the localization of valence band maxima (VBM) occurs at M point and the localization of conduction band minima (CBM) occurs at Γ point. In case of spin down contribution, the electronic band gap is direct in nature, which means the hole crystal momentum and electron crystal momentum are same. The value of the electronic band gap is 4.63 eV, which makes it wide band gap semiconductor. Moreover, the VBM and CBM localize at M point. The difference in the electronic band gap for spin up and spin down contributions can be

observed in the DOS as well. Moreover, for spin up configuration, the contributions of *Cr-3d* state and *Cl-3p* state are in similar fashion, with *Cr-3d* state contributing a little less in the valence band and *Cl-3p* state contributing a little less in the conduction band. However, in the case of spin down configuration, *Cr-3d* state dominates in the conduction band, whereas *Cl-3p* state majorly contributes in the valence band.

It should be noted that the variation in the electronic properties, owing to the alignment of electron spin, indicates the ferromagnetic behaviour of CrCl_3 monolayer. Also, the calculated total magnetization of the stable ferromagnetic structure is $6.0 \mu_B/\text{cell}$. Here, the spin dependent electronic properties and ferromagnetic nature make CrCl_3 monolayer a promising candidate for spintronic devices, which can be great alternative to CMOS devices for energy efficient data storage [16].

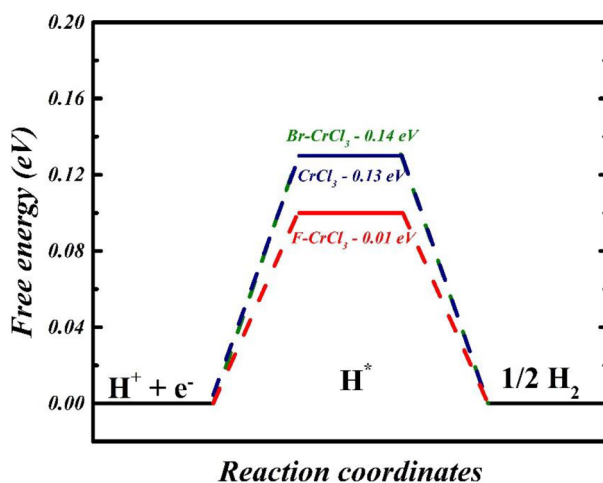
3.3 Hydrogen evolution reaction

Hydrogen evolution reaction (HER) is one of the vital electrochemical processes for photocatalysis. HER is the cathode reaction during the photocatalytic water splitting. HER is a simple process, which is a combination of two steps. One is adsorption of hydrogen by the catalyst surface, which is known as *Volmer* reaction and the other is evolution of hydrogen molecule through, either combining two adsorbed hydrogens or combining the adsorbed hydrogen with the hydrogen of other molecules. The first mentioned process is called *Tafel* reaction and the latter is known as *Heyrovsky* reaction. Hence, the evolution of hydrogen takes place through either *Volmer-Tafel* reaction or *Volmer-Heyrovsky* reaction, depending on the behaviour of the catalyst surface [25, 26].

The efficiency of a catalyst for hydrogen evolution can be computed by verifying the Gibbs free energy of the catalytic surface for hydrogen adsorption. The catalyst is considered the most efficient if the Gibbs free energy is 0 eV. The Gibbs free energy for hydrogen adsorption can be calculated using equation, $\Delta G^0 = \Delta E_{ads} - \Delta E_{ZPE} - T\Delta S_H$. Here, ΔE_{ads} is the adsorption energy for hydrogen adsorption, ΔE_{ZPE} is the zero-point energy of hydrogen and $T\Delta S_H$ is the entropy difference of hydrogen in different phase. In standard conditions, $\Delta E_{ZPE} - T\Delta S_H$ is equals to -0.24 eV. Hence, the Gibbs free energy $\Delta G^0 = \Delta E_{ads} + 0.24$ [27]. In case of pristine CrCl_3 monolayer, the Gibbs free energy for intermediate hydrogen adsorption is 0.13 eV, which indicates not so good efficiency for hydrogen evolution. However, as compared to WS_2 nanosheets and FeP nanoparticles, pristine CrCl_3 possesses better catalytic activity [28, 29].

It has been observed that the activity for hydrogen evolution can be improved by functionalizing the catalytic surface. One of the ways to functionalize the surface is through substitutional doping. In the present work, one Cl atom was substituted by F atom and Br atom, one by one, in order to functionalize the CrCl_3 monolayer. Due to the substitution, a feeble variation was observed in the electronic properties of CrCl_3 monolayer. In case of F-substituted CrCl_3 monolayer, the electronic band gaps became 1.89 eV and 4.58 eV for spin up and spin down configurations, respectively. While for Br-substituted CrCl_3 monolayer, the respective electronic band gaps for spin up and spin down configurations are 1.83 eV and 4.38 eV. Figure 3 depicts the electronic band structure and density of states for F- CrCl_3 and Br- CrCl_3 . Very minor contributions from the substituted halogens (F and Br) can be observed in the projected density of states (PDOS). Moreover, the Gibbs free energy for hydrogen adsorption, in case of F- CrCl_3 reduces to 0.1 eV, whereas for Br- CrCl_3 remains

Fig. 4 Free energy diagram for HER in case of pristine, F substituted and Br substituted CrCl_3 monolayer



0.13 eV, which suggests that fluorine substitution improves the catalytic activity of CrCl_3 monolayer for hydrogen evolution by as much as 23%. Figure 4 depicts the free energy diagram for HER. The variation in the catalytic activity is clearly visible in case F-CrCl_3 monolayer.

4 Conclusions

A density functional theory (DFT) based study, in the framework of the first principles method, of the structural and electronic properties of CrCl_3 monolayer was carried out. Having an octahedral symmetry, the CrCl_3 monolayer attains trigonal crystal structure with lattice parameters $a=b=5.99 \text{ \AA}$ and $\gamma=120^\circ$. The electronic band structure is evocative of semiconducting nature of CrCl_3 monolayer for both spin up and spin down configurations. The calculated electronic band gaps are 2.06 eV and 4.63 eV, for spin up and spin down configurations, respectively. Moreover, from PDOS the dominance of $\text{Cr-}3d$ orbital in conduction band and the dominance of $\text{Cl-}3p$ state in the valence band can be confirmed. With the total magnetization of $6.0 \mu_B/\text{cell}$, the ferromagnetic behaviour of CrCl_3 can be confirmed, making it an option for spintronic devices. Furthermore, the catalytic activity for hydrogen evolution of CrCl_3 was studied by calculating the Gibbs free energy for hydrogen adsorption. A very decent activity for hydrogen evolution was observed for pristine CrCl_3 monolayer with ΔG^0 equal to 0.13 eV. On substitution of fluorine, the ΔG^0 becomes 0.1 eV, indicating enhancement upto 23% in the catalytic activity for hydrogen evolution.

Acknowledgements The computer facility developed under DST-FIST Level-I (No.SR/FST/PSI-097/2006 dated 20th December 2006 and No.SR/FST/PSI-198/2014 dated 21st November 2014) programmes of Department of Science and Technology, Government of India, New Delhi and support under DRS-SAP-I (No. F-530/10/DRS/2010 (SAP-I) dated November 2010 and No.F.530/17/DRS-II/2018 (SAP-II), dated 17th April 2018) of University Grants Commission, New Delhi are highly acknowledged. VBZ would acknowledge the project fellowship under DRS-II-SAP (No.F.530/17/DRS-II/2018 (SAP-I), dated 17/04/2018) of University Grants Commission, New Delhi. RSS would like to thank Council of Scientific and Industrial Research (CSIR), New Delhi for providing with the NET-SRF fellowship (No: 09/070(0069)/2019-EMR-I). SKG would like to thank Science and Engineering Research Board (SERB), India for the financial support (Grant no.: YSS/2015/001269 and CRG/2022/007329). Also, SKG is thankful to the St. Xavier's College,

Ahmedabad for providing computing facility and also financial assistance under the Scheme of Research Project Seed Grant having award nos. SXCA/2021-2022/ PR07 and SXCA/2021-2022/ PR09.

Author contributions VBZ modelled the structure and carried out calculations. VBZ also prepared the first draft of the manuscript. RSS worked with the methodology and computations. RSS reviewed the manuscript. SKG conceptualized the work, provided insights and reviewed the manuscript. PNG conceptualized the work, provided funding and infrastructure for computation and reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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